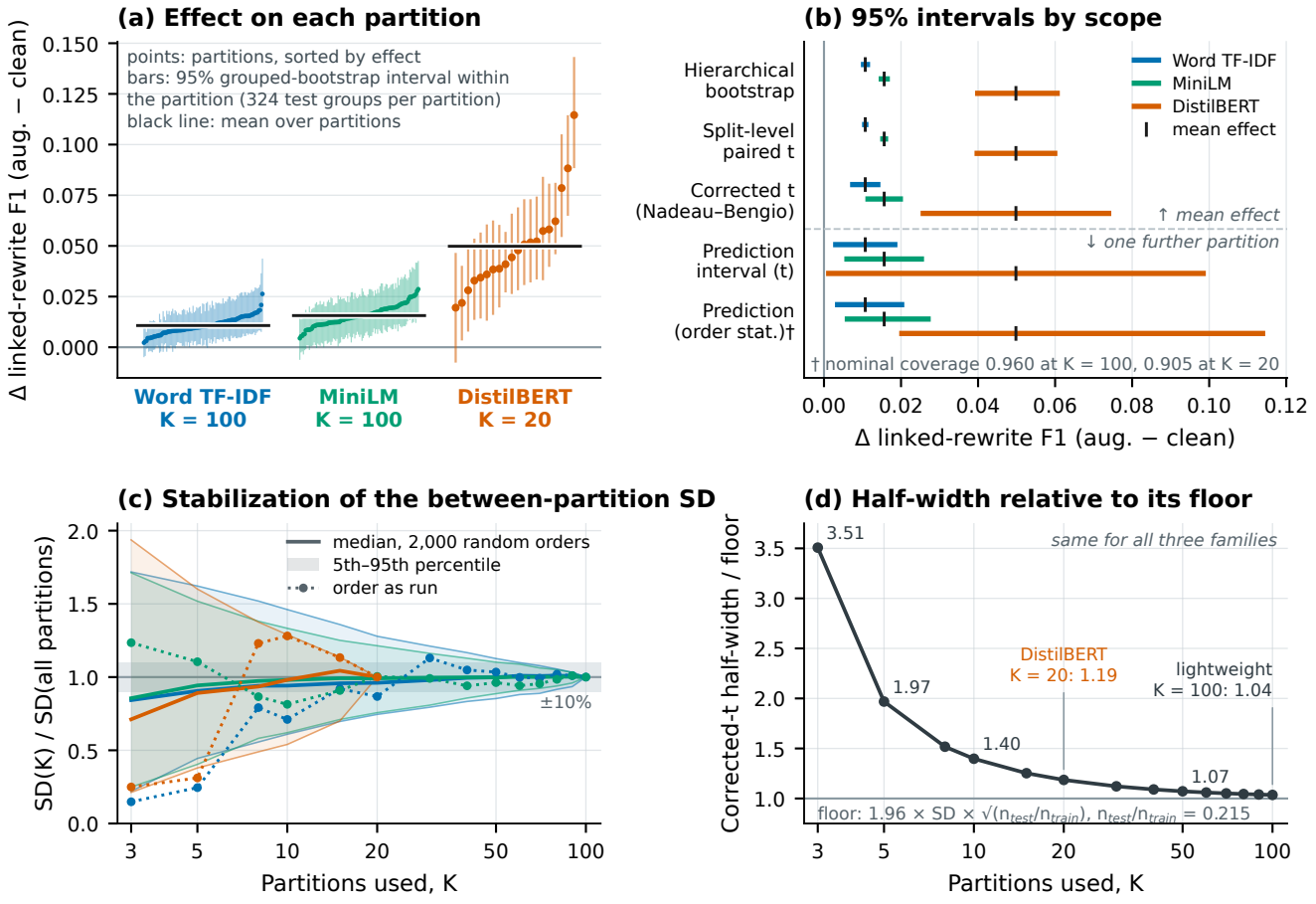
(a) Effect on each partition
points: partitions, sorted by effect
bars: 95% grouped-bootstrap interval within the partition (324 test groups per partition)
black line: mean over partitions
Δ linked-rewrite F1 (aug. − clean)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Word TF-IDF K = 100
MiniLM K = 100
DistilBERT K = 20
(b) 95% intervals by scope
Hierarchical bootstrap
Split-level paired t
Corrected t (Nadeau–Bengio)
Prediction interval (t)
Prediction (order stat.)†
Word TF-IDF
MiniLM
DistilBERT
mean effect
↑ mean effect
↓ one further partition
† nominal coverage 0.960 at K = 100, 0.905 at K = 20
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Δ linked-rewrite F1 (aug. − clean)
(c) Stabilization of the between-partition SD
SD(K) / SD(all partitions)
median, 2,000 random orders
5th–95th percentile
order as run
±10%
2.0
1.5
1.0
0.5
0.0
3
5
10
20
50
100
Partitions used, K
(d) Half-width relative to its floor
Corrected-t half-width / floor
same for all three families
3.51
1.97
1.40
1.07
DistilBERT K = 20: 1.19
lightweight K = 100: 1.04
floor: 1.96 × SD × √(n_test/n_train), n_test/n_train = 0.215
3.5
3.0
2.5
2.0
1.5
1.0
3
5
10
20
50
100
Partitions used, K